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NEXTGEN AI PROJECT REPORT

ON

SERENOVA – AI MENTAL HEALTH COGNITIVE RISK PREDICTOR

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In partial fulfilment for the award of the degree of MCA.

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1. PROBLEM STATEMENT

In recent years, mental health concerns such as stress, anxiety, and emotional burnout have significantly increased among students and young professionals. Existing assessment methods primarily rely on static questionnaires and fail to incorporate behavioural and linguistic indicators that can provide deeper insights into an individual's cognitive state.

There is a need to develop an AI-driven, web-based multi-modal system that integrates psychological responses, sentiment analysis, and behavioural typing patterns to identify early cognitive risk levels and provide supportive, non-diagnostic interventions for improved mental well-being.

2. INTRODUCTION

Mental health risk assessment remains largely dependent on static self-report instruments that often fail to capture dynamic behavioural and linguistic signals associated with cognitive distress. With increasing academic pressure, digital exposure, and social isolation, early detection mechanisms must evolve beyond traditional screening frameworks.

Advancements in Artificial Intelligence, Natural Language Processing, and Behavioural Analytics enable the development of multi-dimensional assessment systems capable of analysing psychological, emotional, and behavioural patterns simultaneously. This project proposes a structured AI-driven web framework for early cognitive risk identification using integrated multi-modal data analysis.

3. OBJECTIVES

- 1. To Architect a Multi-Modal Cognitive Risk Modelling Framework:** Design a unified analytical architecture that integrates psychological survey responses, linguistic sentiment features, and behavioural typing biomarkers into a cohesive predictive system.
- 2. To Enhance Contextual Depth in Mental Health Screening:** Incorporate real-time linguistic and behavioural signals to complement traditional self-report measures, thereby improving contextual understanding of emotional and cognitive states.
- 3. To Develop a Probabilistic Risk Classification Model:** Implement supervised machine learning techniques to generate interpretable risk probabilities (Low, Moderate, High) based on fused multi-source features.
- 4. To Ensure Explainability and Model Transparency:** Integrate explainable AI principles to quantify feature contribution, ensuring analytical interpretability and increasing user trust.
- 5. To Establish an Ethical, Non-Diagnostic Support Mechanism:** Design a responsible AI intervention layer that delivers awareness-oriented guidance while maintaining strict privacy, data protection, and non-clinical boundaries.

4. METHODOLOGY

Phase 1: Data Acquisition

- Curated mental health survey datasets for supervised risk modelling
- Sentiment-labeled textual datasets for NLP training
- Real-time user-generated questionnaire responses, text inputs, and typing metrics

Phase 2: Data Preprocessing & Feature Engineering

- Numerical encoding of psychological assessment variables
- NLP preprocessing: tokenization, lemmatization, TF-IDF/vector embeddings
- Extraction of behavioral features: typing speed variance, pause latency, correction frequency
- Feature normalization and scaling

Phase 3: Multi-Modal Feature Fusion

- Construct a composite feature vector integrating:
- Psychological score
- Sentiment polarity & emotional intensity
- behavioural typing indicators

Phase 4: Predictive Modelling

- Supervised learning (Logistic Regression / Random Forest)
- Probabilistic risk scoring
- Cross-validation for performance optimization

Phase 5: Deployment Architecture

- Web Interface (interactive input & dashboard)
- Backend ML inference engine
- Secure database management
- Visualization layer for interpretability

5. PROPOSED SOLUTION

- Develop a **multi-modal AI-based web platform** integrating psychological, linguistic, behavioural, and optional visual indicators for early cognitive risk detection.
- Implement **sentiment and emotion analysis** on user-generated text using NLP techniques to extract emotional polarity and intensity.
- Analyse **behavioural typing patterns** (speed variation, pause latency, correction frequency) as cognitive biomarkers.
- Apply **machine learning-based risk classification models** to generate probabilistic cognitive risk levels (Low / Moderate / High).
- Provide an **explainable, non-diagnostic support system** offering personalized positive recommendations while ensuring privacy and ethical compliance.